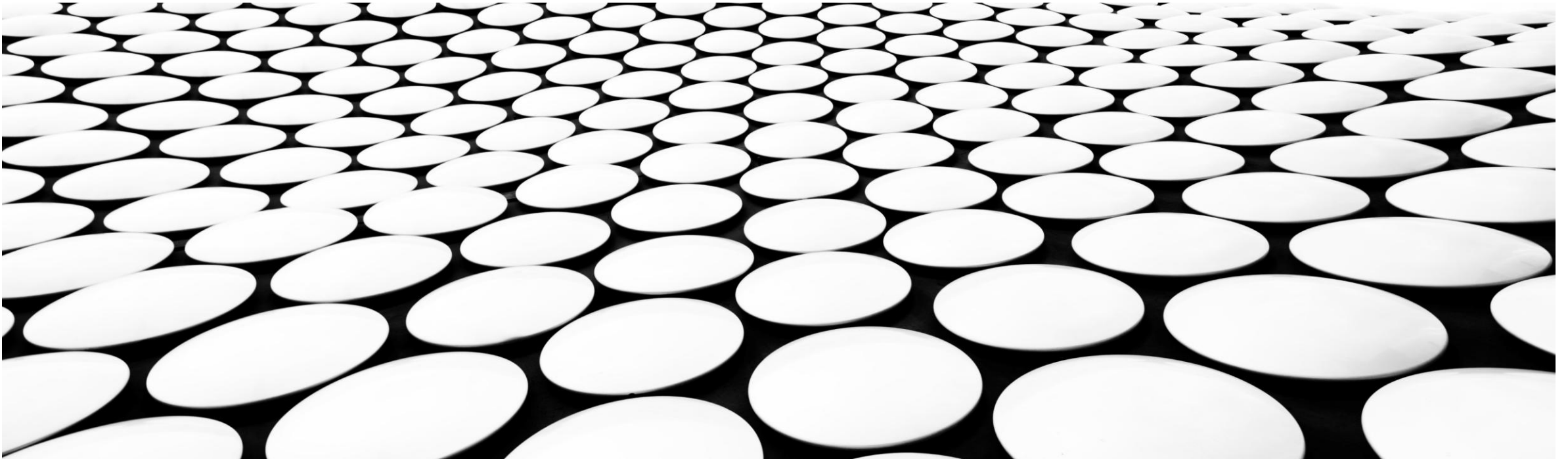
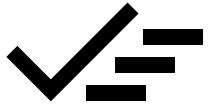


NAIVE BAYES CLASSIFIER

Introduction and Example





NAIVE BAYES CLASSIFIER

- Naives Bayes for Classification
- Conditional Independence Assumption
- Analysis

$$P(\text{instrumental} = \text{Yes} \mid \text{Jazz}) P(\text{Jazz})$$

$$= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(\text{instrumental} = \text{Yes} \mid \text{Folk}) P(\text{Folk})$$

$$= \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{6}$$

$$P(\text{instrumental} = \text{Yes} \mid \text{Rock}) P(\text{Rock})$$

$$= \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

Naives Bayes Classification

Probabilities

Event A → 2-dice rolls providing a sum of 6

{1,5}, {2,4}, {3,3}, {4,2}, {5,1}

Event B → 2-dice rolls where the outcome of one dice is twice the outcome of the other dice

{1,2}, {2,1}, {2,4}, {4,2}, {3,6}, {6,3}



Prior Probability - Probability of the Event

$$P(A) = 5/36$$

$$P(B) = 6/36$$

Joint Probability - Probability of both events occurring together

$$p(A \cap B) = 2/36$$

Conditional Probability – Probability of an event provided the other event already happened

$$P(A|B) = p(A \cap B)/p(B) = (2/36) / (6/36) = 1/3$$

Baye's Rule

$$p(A|B) = p(A \cap B) / p(B) \quad (\text{eq. 1})$$

$$p(B|A) = p(A \cap B) / p(A) \quad (\text{eq. 2})$$

Rewrite equation 2 above:

$$p(A \cap B) = p(B|A) * p(A) \quad (\text{eq. 3})$$

Insert equation 3 in equation 1

$$p(A|B) = p(B|A) * p(A) / p(B) \quad (\text{eq. 4})$$

$$p(A|B) = p(B|A) * p(A) / p(B)$$



Bayes Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Hypothesis Testing

Naïve Bayes for classification

**Assume there are individual hypotheses, h_i , from a set of hypotheses, H .
Assume a set of evidences, E .**

$$p(h_i|E) = (p(E|h_i) * p(h_i)) / p(E)$$

This equation may be read, "**The probability of an hypothesis h_i given a set of evidences E is . . .**"

$$P(h_i|E) = \frac{P(E|h_i)P(h_i)}{P(E)}$$

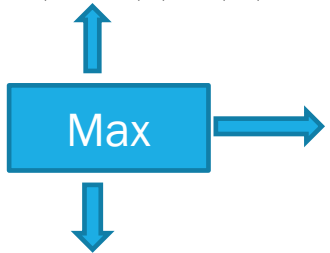
Hypothesis is the class

Evidence is the set of feature values

Example

Two hypotheses: Bike or Drive

$$p(\text{Bike}|E) = (p(E|\text{Bike}) * p(\text{Bike})) / p(E)$$



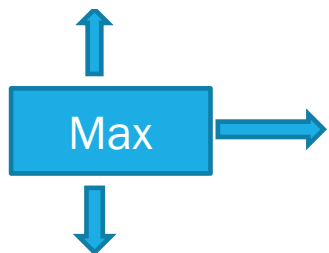
$$p(\text{Drive}|E) = (p(E|\text{Drive}) * p(\text{Drive})) / p(E)$$

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

Example

Two hypotheses: Bike or Drive

$$p(\text{Bike}|E) \propto p(E|\text{Bike}) * p(\text{Bike})$$



$$p(\text{Drive}|E) \propto p(E|\text{Drive}) * p(\text{Drive})$$

Priors

$$p(\text{Bike}) = 9/15$$

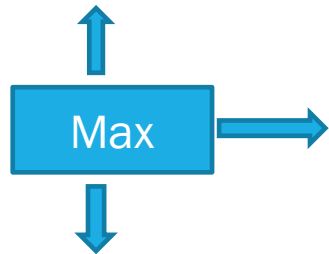
$$P(\text{Drive}) = 6/15$$

Very sensitive to
class imbalance

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

Suppose that the event E is (T=Average)

$$\begin{aligned} P(\text{Bike}|\text{Average}) &\propto P(\text{Average}|\text{Bike}) * P(\text{Bike}) \\ &\propto 2/9 * 9/15 \\ &\propto 0.13 \end{aligned}$$



$$\begin{aligned} P(\text{Drive}|\text{Average}) &\propto P(\text{Average}|\text{Drive}) P(\text{Drive}) \\ &\propto 1/6 * 6/15 \\ &\propto 0.06 \end{aligned}$$

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

Independence Assumption

Suppose that event E is (T=Average AND R=Lot)

The « naive » part of Naive Bayes...
considering features as independent

Conditionally Independent Events

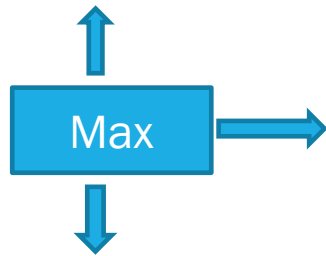
Two events A and B are conditionally independent of each other given a third event C if and only if:

$$p((A \cap B)|C) = p(A|C) * p(B|C)$$

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

Suppose that the event is $E = (T=\text{Average AND } R=\text{Lot})$

$$\begin{aligned} P(\text{Bike} | (\text{Average \& Lot})) &\propto P(\text{Average \& Lot} | \text{Bike}) P(\text{Bike}) \\ &\propto P(\text{Average} | \text{Bike}) * P(\text{Lot} | \text{Bike}) * P(\text{Bike}) \\ &\propto 2/9 * 1/9 * 9/15 \\ &\propto 0.015 \end{aligned}$$



$$\begin{aligned} P(\text{Drive} | (\text{Average \& Lot})) &\propto P(\text{Average \& Lot} | \text{Drive}) P(\text{Drive}) \\ &\propto P(\text{Average} | \text{Drive}) * P(\text{Lot} | \text{Drive}) * P(\text{Drive}) \\ &\propto 1/6 * 2/6 * 6/15 \\ &\propto 0.022 \end{aligned}$$

Sample	Temperature	Rain/snow	Bike / Drive
s1	Very Cold	Normal	Drive
s2	Very Cold	Normal	Drive
s3	Very Cold	Normal	Drive
s4	Cold	Normal	Drive
s5	Cold	Lot	Drive
s6	Average	Lot	Drive
s7	Cold	Little	Bike
s8	Average	Little	Bike
s9	Warm	Little	Bike
s10	Average	Little	Bike
s11	Warm	Normal	Bike
s12	Warm	Normal	Bike
s13	Warm	Lot	Bike
s14	Warm	Little	Bike
s15	Warm	Normal	Bike

9

Analysis

Model learning

Nothing is « created » except a set of prior and posterior probabilities

Phone ID	Defective (Y)	Sensor Score (X_1)	Assembly Line (X_2)	Worker Experience (X_3)
001	1 (Yes)	45	Line A	Junior
002	0 (No)	90	Line B	Senior
003	1 (Yes)	50	Line C	Junior
004	0 (No)	85	Line A	Mid
005	1 (Yes)	55	Line B	Junior
006	0 (No)	95	Line C	Senior

Works with Categorical features, requires binning

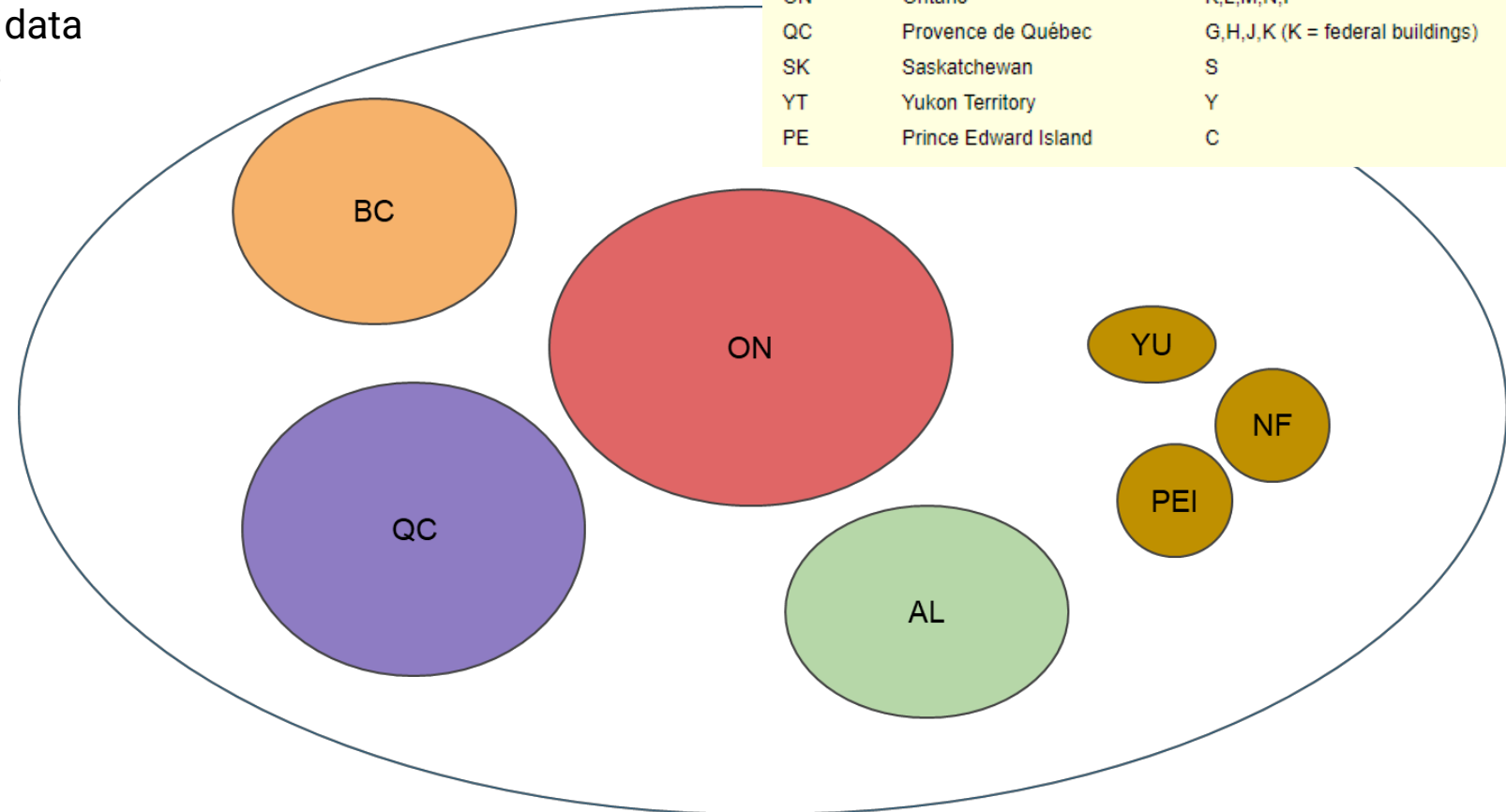
Algorithmic Bias

The learner assumes the data comes from more or less probable classes.

■ 1N 6T5

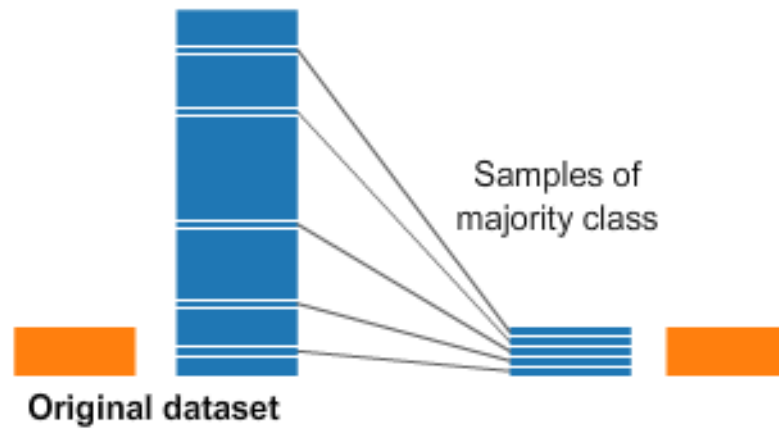
O1N 6T5

Code	Province name	First letter of postal code
AB	Alberta	T
BC	British Columbia	V
MB	Manitoba	R
NB	New Brunswick	E
NL	Newfoundland & Labrador	A
NS	Nova Scotia	B
NT	Northwest Territories	X
NU	Nunavut	X
ON	Ontario	K,L,M,N,P
QC	Province de Québec	G,H,J,K (K = federal buildings)
SK	Saskatchewan	S
YT	Yukon Territory	Y
PE	Prince Edward Island	C



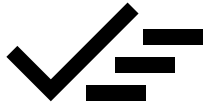
Class imbalance

Undersampling



Oversampling





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